

Détection d'une attaque Brute Force avec Wazuh

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Configuration du serveur Wazuh

Après l'installation de la machine virtuelle Wazuh, nous démarrons le serveur.

[illegible]

FIGURE 1.1 – Démarrage du serveur Wazuh

Chapitre 2

Vérification de l'adresse IP

Pour connaître l'adresse IP du serveur nous utilisons :

`ip a`

```
[wazuh-user@wazuh-server ~]$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:92:29:34 brd ff:ff:ff:ff:ff:ff
    altname enp0s17
    inet 10.53.21.30/24 metric 1024 brd 10.53.21.255 scope global dynamic eth0
        valid_lft 3573sec preferred_lft 3573sec
    inet6 fe80::a00:27ff:fe92:2934/64 scope link proto kernel_ll
        valid_lft forever preferred_lft forever
[wazuh-user@wazuh-server ~]$
```

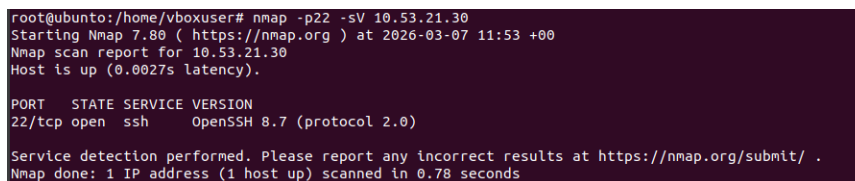
FIGURE 2.1 – Adresse IP du serveur Wazuh

Chapitre 3

Test du service SSH

Nous vérifions que le service SSH est actif en utilisant Nmap.

```
nmap -p22 -sV 192.168.1.13
```



```
root@ubuntu:/home/vboxuser# nmap -p22 -sV 10.53.21.30
Starting Nmap 7.80 ( https://nmap.org ) at 2026-03-07 11:53 +00
Nmap scan report for 10.53.21.30
Host is up (0.0027s latency).

PORT      STATE SERVICE VERSION
22/tcp    open  ssh      OpenSSH 8.7 (protocol 2.0)

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 0.78 seconds
```

FIGURE 3.1 – Scan du port SSH

Connexion SSH

```
ssh wazuh-user@192.168.1.13
```

[illegible]

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Chapitre 5

Accès au Dashboard Wazuh

Nous accédons au dashboard via le navigateur en utilisant l'adresse IP du serveur.

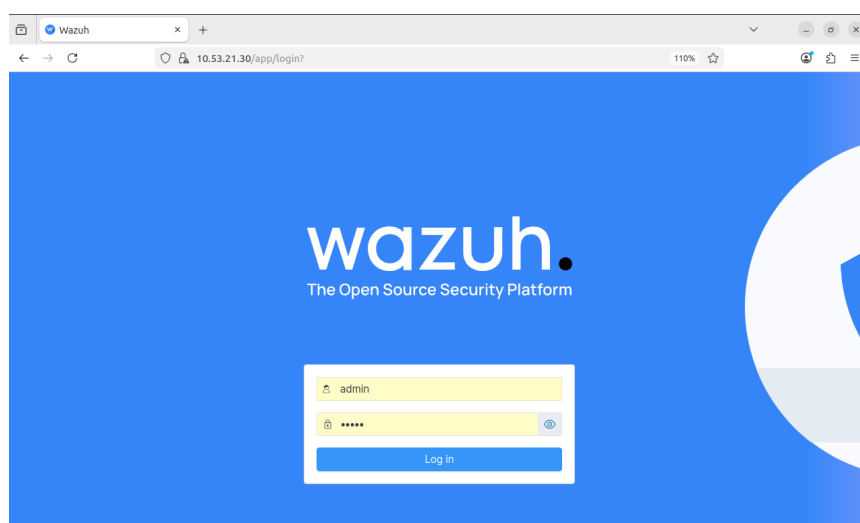


FIGURE 5.1 – Dashboard Wazuh

Chapitre 6

Installation du Wazuh Agent

Installation de l'agent sur la machine cible.

```
walid@walid-VirtualBox:~$ wget https://packages.wazuh.com/4.x/apt/pool/main/w/wazuh-agent/wazuh-agent_4.14.3-1_and64.deb && sudo WA
UH_MANAGER='10.53.21.30' WAZUH_AGENT_NAME='testtest' dpkg -i ./wazuh-agent_4.14.3-1_and64.deb
--2026-03-07 07:20:33-- https://packages.wazuh.com/4.x/apt/pool/main/w/wazuh-agent/wazuh-agent_4.14.3-1_and64.deb
Resolving packages.wazuh.com (packages.wazuh.com)... 99.84.9.57, 99.84.9.62, 99.84.9.92, ...
Connecting to packages.wazuh.com (packages.wazuh.com)[99.84.9.57]:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 13213254 (13M) [application/vnd.debian.binary-package]
Saving to: 'wazuh-agent_4.14.3-1_and64.deb'

wazuh-agent_4.14.3-1_and64.deb 100%[=====] 12.60M 1.44MB/s in 7.2s

2026-03-07 07:20:40 (1.76 MB/s) - 'wazuh-agent_4.14.3-1_and64.deb' saved [13213254/13213254]

[sudo] password for walid:
Selecting previously unselected package wazuh-agent.
(Reading database ... 199963 files and directories currently installed.)
Preparing to unpack .../wazuh-agent_4.14.3-1_and64.deb ...
Unpacking wazuh-agent (4.14.3-1) ...
Setting up wazuh-agent (4.14.3-1) ...
```

FIGURE 6.1 – Installation du wazuh-agent

Chapitre 7

Activation de l'agent

```
sudo systemctl enable wazuh-agent  
sudo systemctl start wazuh-agent
```

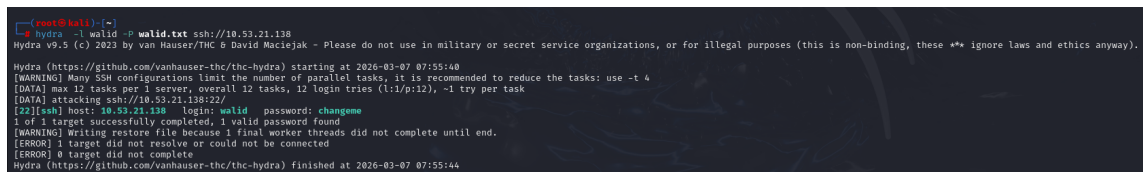
```
walid@walid-VirtualBox: $ systemctl status wazuh-agent  
○ wazuh-agent.service - Wazuh agent  
   Loaded: loaded (/lib/systemd/system/wazuh-agent.service; disabled; vendor preset: enabled)  
   Active: inactive (dead)  
walid@walid-VirtualBox: $ sudo systemctl daemon-reload  
walid@walid-VirtualBox: $ sudo systemctl daemon-reloadchangen  
Unknown command verb daemon-reloadchangen  
walid@walid-VirtualBox: $ sudo systemctl enable wazuh-agent  
Created symlink /etc/systemd/system/multi-user.target.wants/wazuh-agent.service → /lib/systemd/system/wazuh-agent.service.  
walid@walid-VirtualBox: $ sudo systemctl start wazuh-agent  
walid@walid-VirtualBox: $ systemctl status wazuh-agent  
● wazuh-agent.service - Wazuh agent  
   Loaded: loaded (/lib/systemd/system/wazuh-agent.service; enabled; vendor preset: enabled)  
   Active: active (running) since Sat 2026-03-07 07:24:04 EST; 7s ago  
     Process: 4414 ExecStart=/usr/bin/env /var/ossec/bin/wazuh-control start (code=exited, status=0/SUCCESS)  
    Tasks: 29 (limit: 4002)  
   Memory: 28.3M  
      CPU: 2.189s  
   CGroup: /system.slice/wazuh-agent.service  
           └─4436 /var/ossec/bin/wazuh-execd  
             └─4444 /var/ossec/bin/wazuh-agentd  
               └─4457 /var/ossec/bin/wazuh-syscheckd  
                 └─4468 /var/ossec/bin/wazuh-logcollector  
                   └─4484 /var/ossec/bin/wazuh-modulesd  
  
Mar 07 07:24:00 walid-VirtualBox systemd[1]: Starting Wazuh agent...  
Mar 07 07:24:00 walid-VirtualBox env[4414]: Starting Wazuh v4.14.3...  
Mar 07 07:24:00 walid-VirtualBox env[4414]: Started wazuh-execd...  
Mar 07 07:24:01 walid-VirtualBox env[4414]: Started wazuh-agentd...  
Mar 07 07:24:01 walid-VirtualBox env[4414]: Started wazuh-syscheckd...  
Mar 07 07:24:02 walid-VirtualBox env[4414]: Started wazuh-logcollector...  
Mar 07 07:24:02 walid-VirtualBox env[4414]: Started wazuh-modulesd...  
Mar 07 07:24:04 walid-VirtualBox env[4414]: Completed.  
Mar 07 07:24:04 walid-VirtualBox systemd[1]: Started Wazuh agent.
```

FIGURE 7.1 – Activation du wazuh-agent

Chapitre 8

Simulation d'une attaque Brute Force

```
hydra -l walid -P walid.txt ssh://10.53.21.138
```



```
(root@kali) ~  
# hydra -l walid -P walid.txt ssh://10.53.21.138  
Hydra v9.5 (c) 2023 by van Hauser/THC & David Maciejak - Please do not use in military or secret service organizations, or for illegal purposes (this is non-binding, these *** ignore laws and ethics anyway).  
  
Hydra (https://github.com/vanhauser-thc/thc-hydra) starting at 2026-03-07 07:55:40  
[WARNING] Many SSH configurations limit the number of parallel tasks, it is recommended to reduce the tasks: use -t 4  
[DATA] max 12 tasks per 1 server, overall 12 tasks, 12 login tries (l:l/p:12), ~1 try per task  
[DATA] attacking ssh://10.53.21.138:22/  
[22][ssh] host: 10.53.21.138  login: walid  password: changeme  
1 of 1 target successfully completed, 1 valid password found  
[WARNING] writing restore file because 1 final worker threads did not complete until end.  
[ERROR] 1 target did not resolve or could not be connected  
[ERROR] 0 target did not complete  
Hydra (https://github.com/vanhauser-thc/thc-hydra) finished at 2026-03-07 07:55:44
```

FIGURE 8.1 – Attaque brute force avec Hydra

Chapitre 9

Analyse dans le Dashboard Wazuh

Après l'exécution de l'attaque brute force, nous observons les résultats dans le tableau de bord Wazuh.

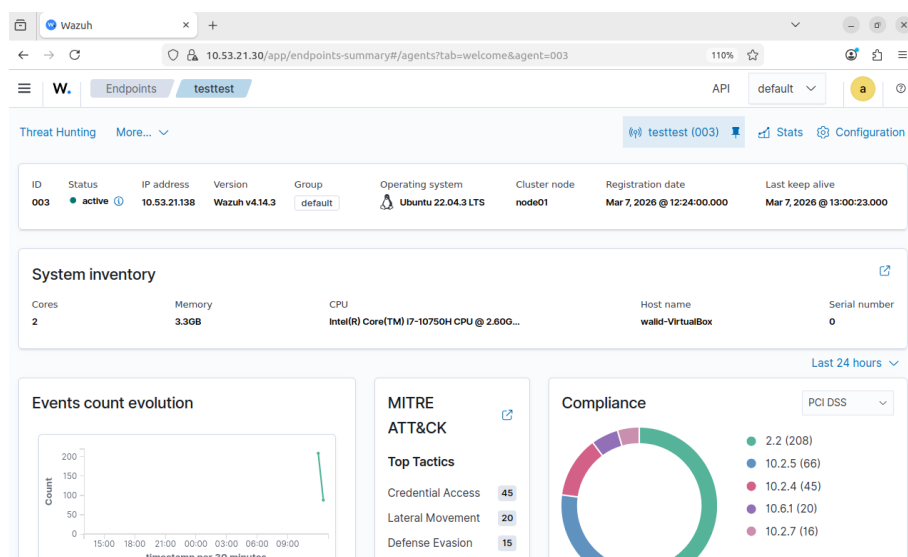


FIGURE 9.1 – Dashboard Wazuh après l'attaque brute force

Dans ce tableau de bord, nous pouvons voir que l'agent est actif et que les événements générés par les tentatives de connexion sont analysés par Wazuh.